Apollonian Circle Packings and Binary Quadratic Forms over $\mathcal{O}_K$

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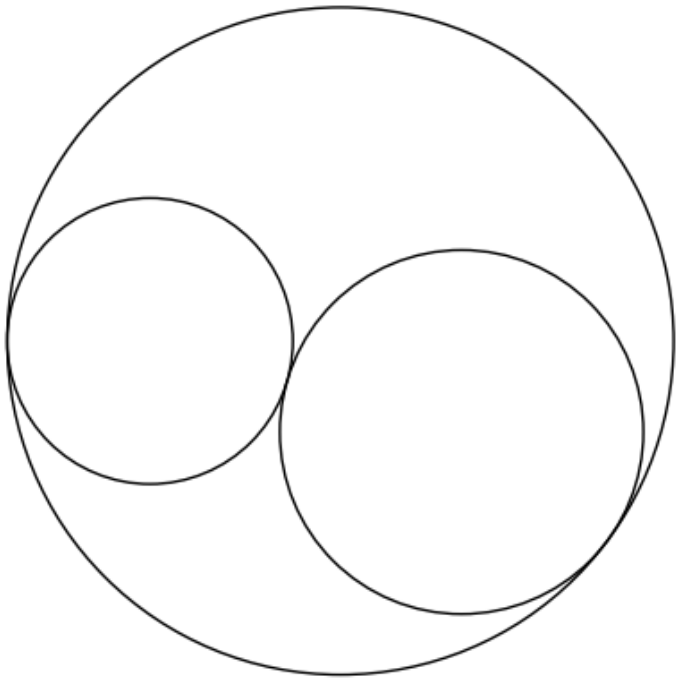
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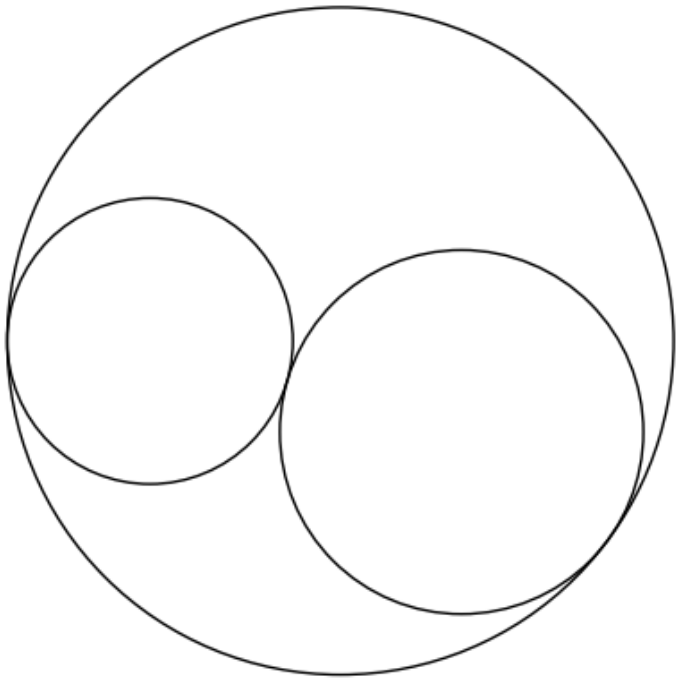
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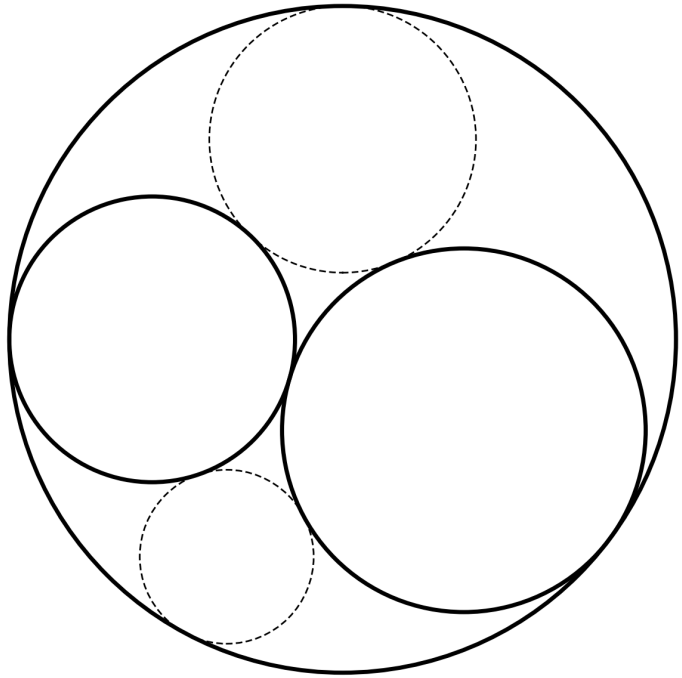
- What is an (integral) Apollonian Circle Packing?
- How do they relate to (integral)Binary Quadratic Forms?
- The relationship with the class group and equivalences of BQFs and ACPs
- Can we generalize from integral to over $\mathcal{O}_K$?

Apollonian Circle Packings

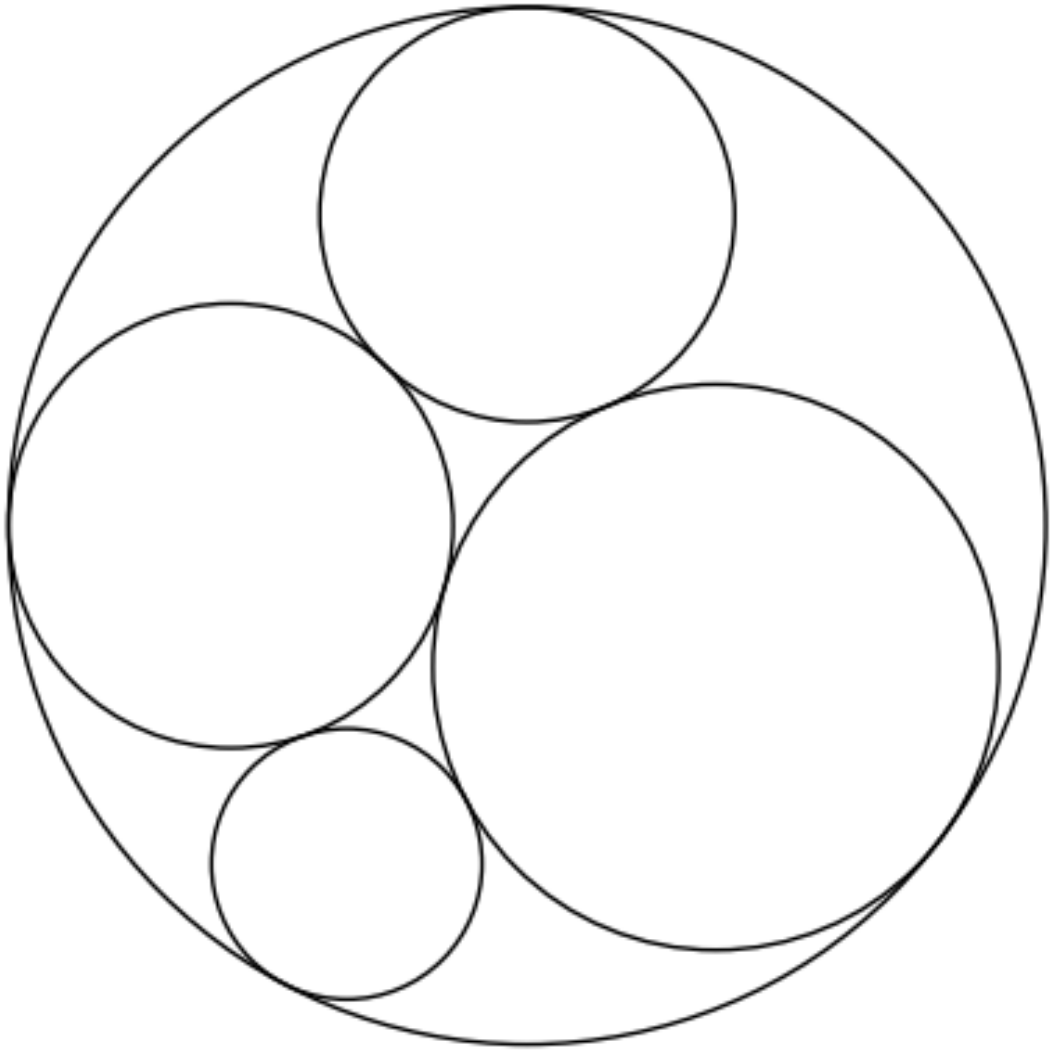


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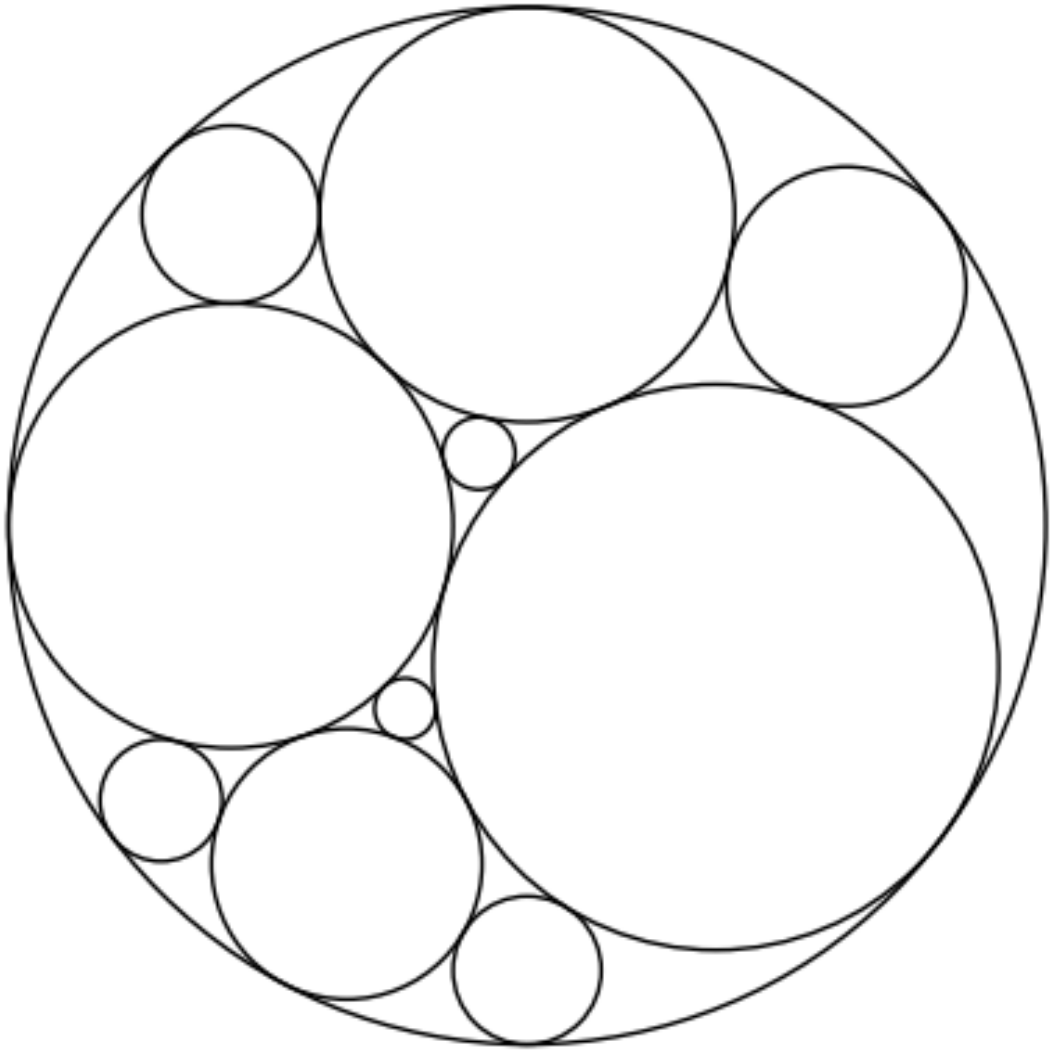




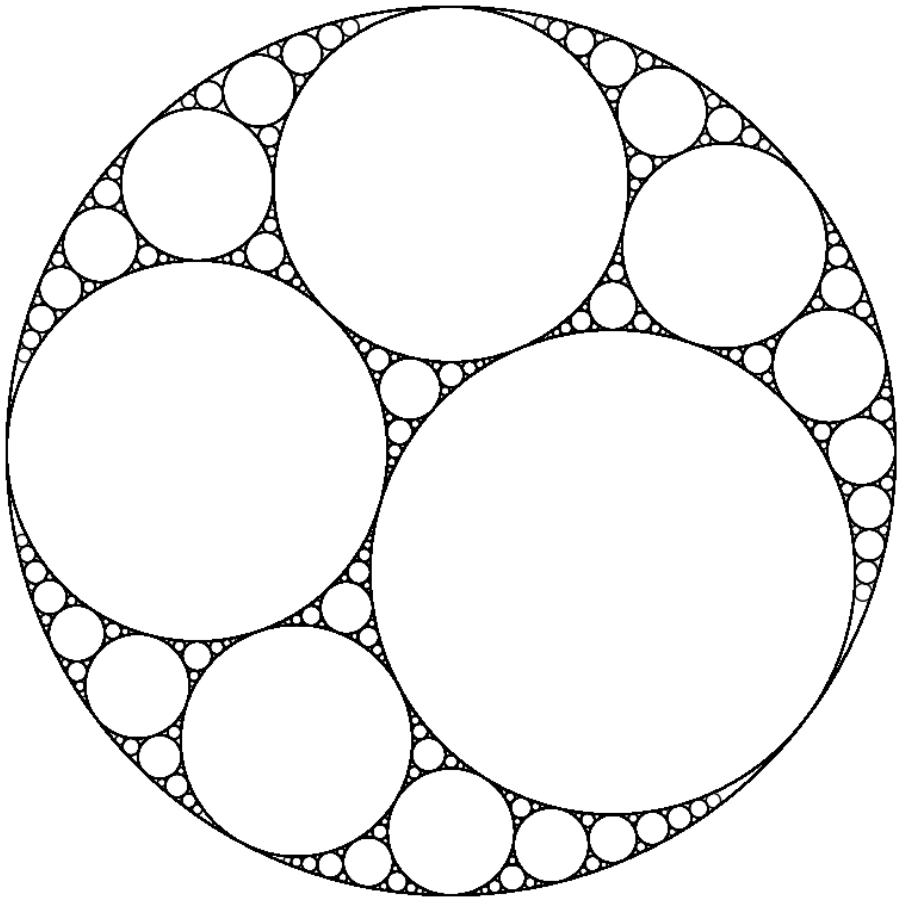
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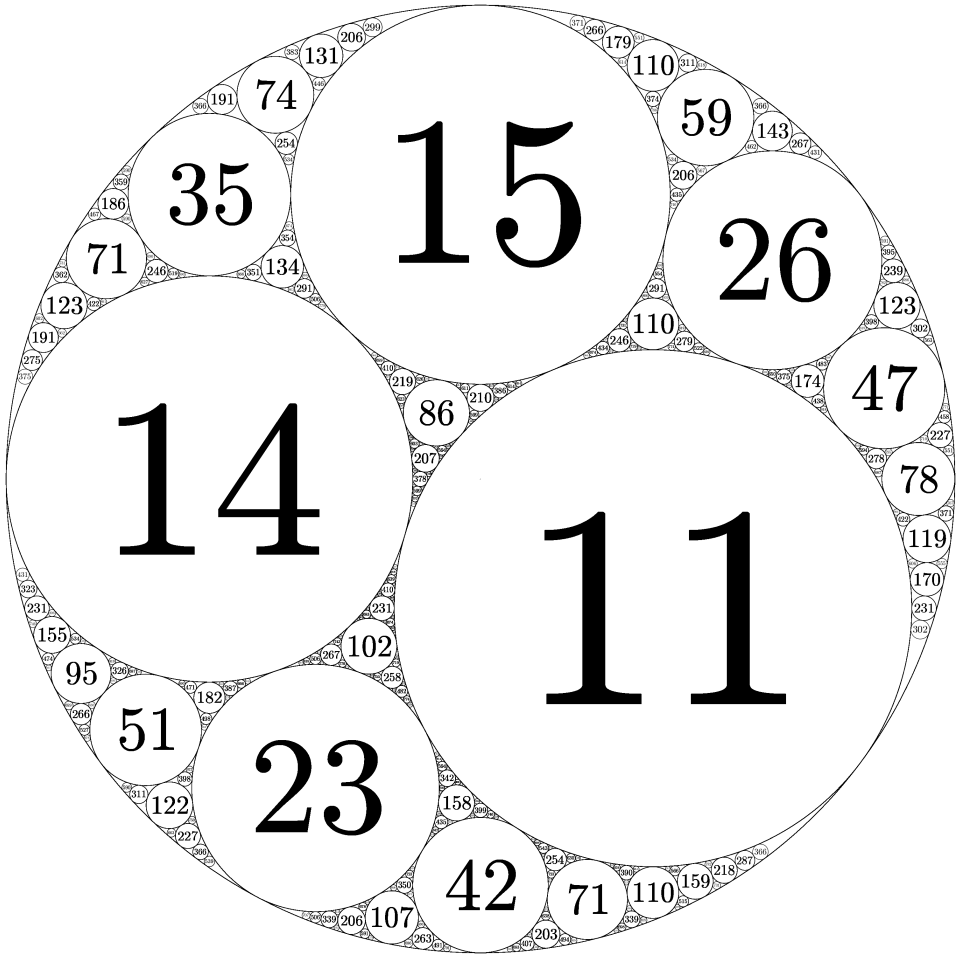
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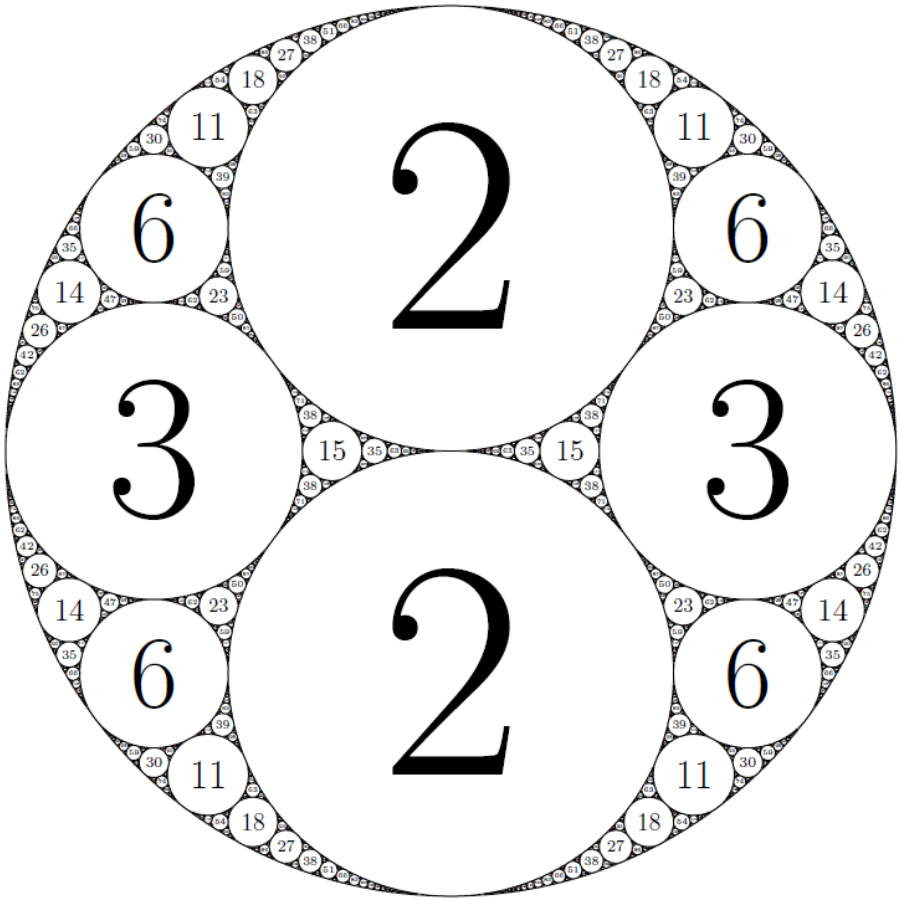
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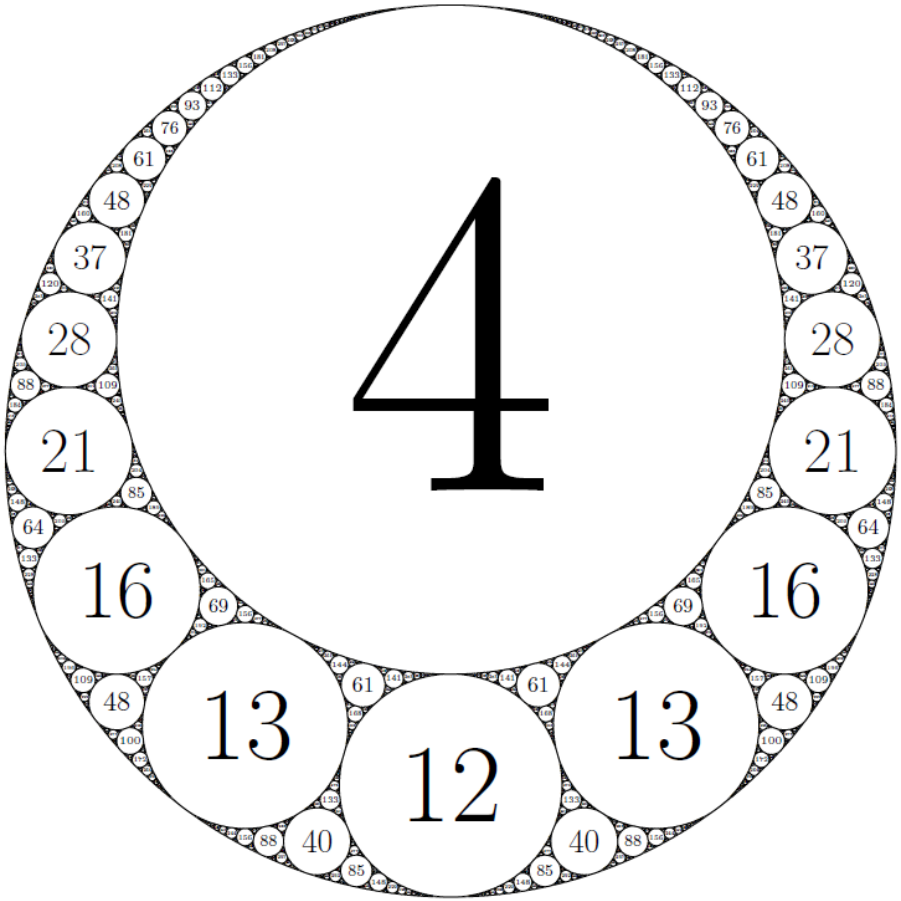


Apollonian Circle Packings



$[-1, 2, 2, 3]$

Apollonian Circle Packings



$[-3, 4, 12, 13]$

Theorem of Elizabeth and Descartes

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Curvatures a, b, c, d of four mutually tangent circles (a *Descartes Quadruple*) then

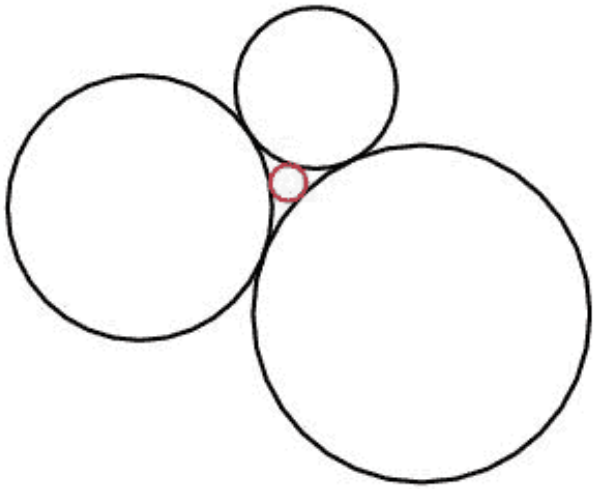
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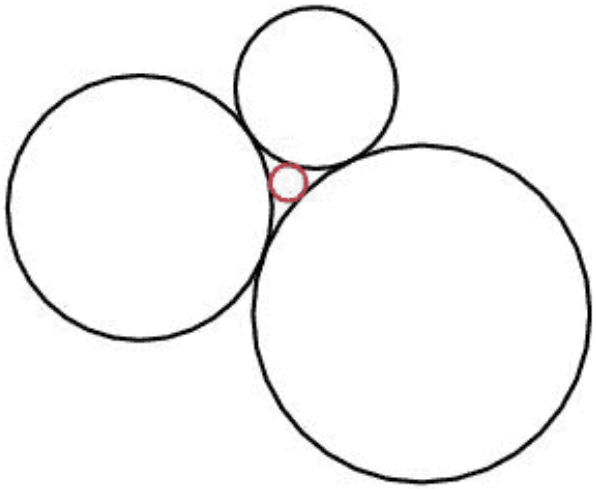
$$(a+b+c+d)^2 = 2(a^2+b^2+c^2+d^2).$$

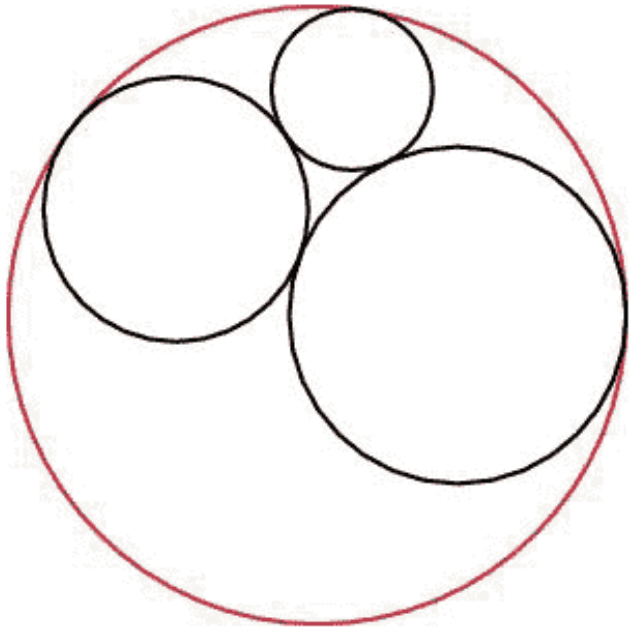
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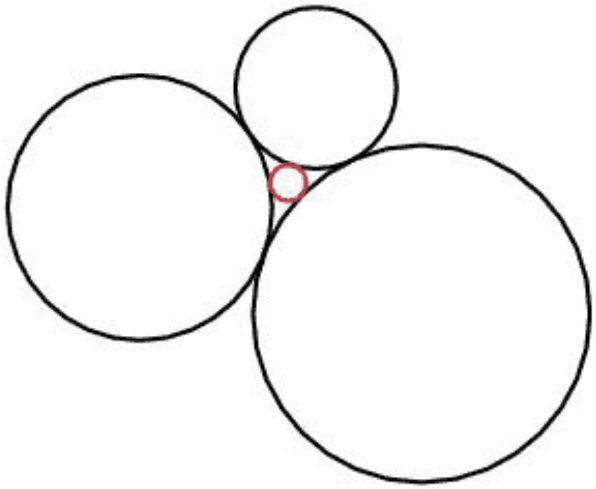


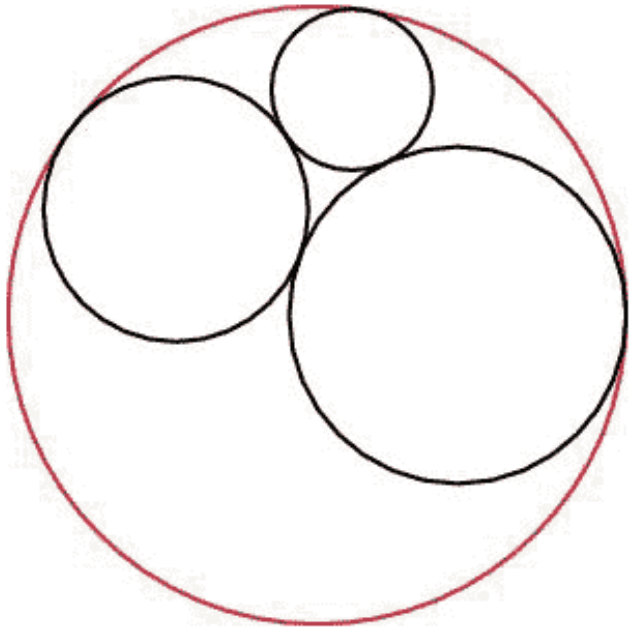
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The two red circles are called *Soddy Circles*

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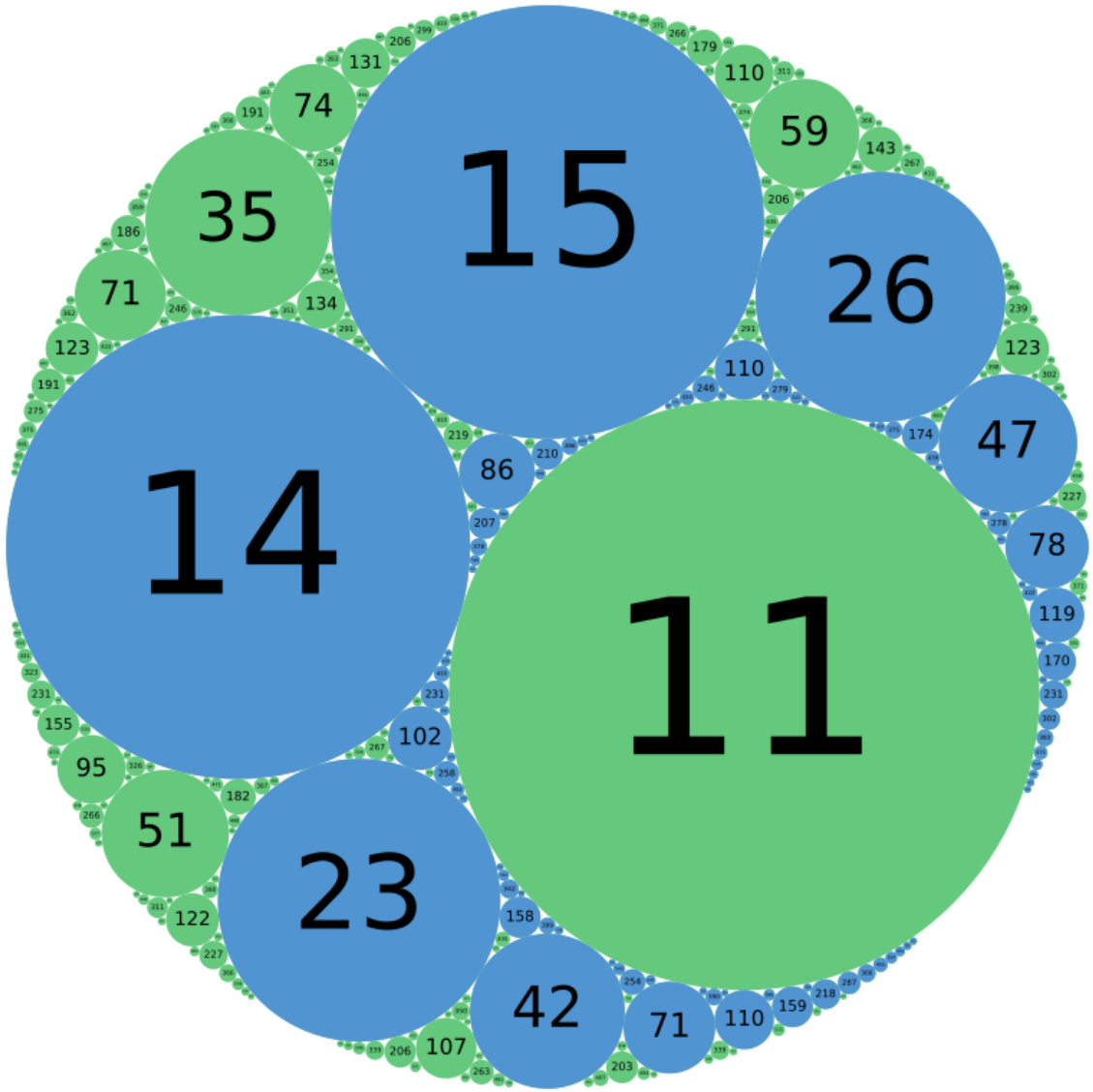
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So if $a, b, c, d \in \mathbb{Z}$ then *curvatures* $\in \mathbb{Z}$

or more generally if $a, b, c, d \in \mathcal{O}_K$ then *curvatures* $\in \mathcal{O}_K$

Let's fix a circle



Why fix a circle?

We get a bijection!

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$$\left\{\begin{array}{c}\text{curv tangent to a fixed} \\ \text{circle with curvature } n\end{array}\right\} \to \left\{\begin{array}{c} Q(x,y) - n : x, y \in \mathbb{Z} \\ \gcd(x,y) = 1 \end{array}\right\}$$

where Q is a BQF with discriminant $-4n^2$

Binary Quadratic Forms

Def: A *Binary quadratic form* (BQF) is

$$\begin{aligned} Q(x,y) &= Ax^2 + Bxy + Cy^2 \\ &= \begin{pmatrix} x & y \end{pmatrix} \begin{pmatrix} A & B/2 \\ B/2 & C \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} \\ &= [A, B, C] \end{aligned}$$

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Def: $\mathrm{PGL}_2(\mathbb{Z})$ acts on BQFs by

$$\begin{aligned} MQ(x,y) &= Q(px + qy, rx + sy) \\ &= \begin{pmatrix} x & y \end{pmatrix} M^T \begin{pmatrix} A & B/2 \\ B/2 & C \end{pmatrix} M \begin{pmatrix} x \\ y \end{pmatrix} \end{aligned}$$

where $M = \begin{pmatrix} p & q \\ r & s \end{pmatrix}$.

BQF actions

Def: Two BQFs, Q, Q' are equivalent if there is some $M \in \mathrm{PGL}_2(\mathbb{Z})$ such that $MQ(x, y) = Q'(x, y)$

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- this action preserves the discriminant and positive definiteness and so divvies up into equivalence classes
- so we're interested in the equivalence class of discriminant of $-4n^2$

Bijection between BQFs and ideals

$$\mathcal{Q}(D) \to Cl^+(\mathcal{O}_D)$$

$$[A, B, C] \mapsto \left[A, \frac{-B + \sqrt{D}}{2}\right]$$

$$\frac{N(\omega_1 x - \omega_2 y)}{N(\mathfrak{a})} \leftarrow \mathfrak{a} = [\omega_1, \omega_2]_{\mathbb{Z}}$$

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$\mathcal{O}_D$ is not necessarily a maximal order so we can use it with $\mathcal{Q}(-4n^2)$ and the order $\mathbb{Z}[ni] \subset \mathbb{Z}[i]$ which is nice

BQFs over Quadratic extensions

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- so the bijection is morally the same but it is only with $Cl(\mathcal{O}_L)$ not just any order

ACPs over $\mathcal{O}_K$

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- So lets take $K = \mathbb{Q}(\sqrt{d})$ and look at ACPs with curvatures in $\mathcal{O}_K$
- there's a LOT more here so we go to a corresponding BQF that has discriminant $-4n^2$ with $n \in \mathcal{O}_K, n \neq 0$

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- in order to keep some of the equivalences* of ACPs we slightly change our setting

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- Let $K = \mathbb{Q}(\sqrt{d})$, $d > 0$ squarefree, and $L = K(i)$

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- this is in general **not** the maximal order $\mathcal{O}_L$
- Mastropietro and Zemková only ever work with $\mathcal{O}_L$, where every ideal is automatically invertible

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- $\mathcal{O}_n$ is quadratic over $\mathcal{O}_K$, not over $\mathbb{Z}$
- that test is no longer available :(

An invertibility criterion

Theorem: For an order $\mathcal{O} \subset L$, every fractional $\mathcal{O}$-ideal is invertible exactly when its multiplier ring is $\mathcal{O}$ iff the $\mathbb{Z}$-dual $\mathcal{O}^\vee$ is invertible

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So for a fractional $\mathcal{O}_n$-ideal $\mathfrak{b}$:

$$\mathfrak{b} \text{ invertible} \iff \{x \in L : x\mathfrak{b} \subset \mathfrak{b}\} = \mathcal{O}_n$$

From a BQF to an ideal

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Proposition: $\mathfrak{a}$ is an ideal of $\mathcal{O}_n$ and $\{x \in L : x\mathfrak{a} \subset \mathfrak{a}\} = \mathcal{O}_n$, so $\mathfrak{a}$ is invertible

Measuring ideals: the norm

For an ideal $I = [\alpha, \beta]_{\mathcal{O}_K}$ of $\mathcal{O}_n$, define the *change of basis matrix* $M_I \in M_2(\mathcal{O}_K)$ by

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Proposition: $N_{\mathcal{O}_n/\mathcal{O}_K}(I) = (\det(M_I))$

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The bijection

$$\psi : Q_n \to \mathrm{Pic}(\mathcal{O}_n)/\sim, \qquad [A, B, C] \longmapsto A\mathcal{O}_K + \left(\frac{-B}{2} + ni\right)\mathcal{O}_K$$

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Theorem: ψ and ϕ are mutually inverse, so

$$Q_n \longleftrightarrow \mathrm{Pic}(\mathcal{O}_n)/\sim$$

Back to ACPs

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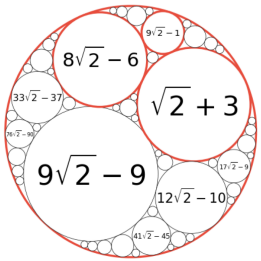
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- lets look at the geometry!

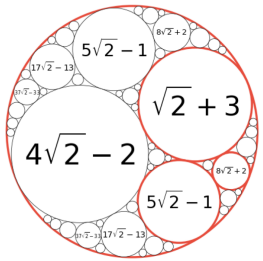
- Quadruple: $(-2, 5\sqrt{2}-3, \sqrt{2}+3, 6\sqrt{2}-4)$
- BQF:$[5\sqrt{2}-5, 2, \sqrt{2}+1]$
- ideal: $(5\sqrt{2}-5)\mathcal{O}_K + (-1-2i)\mathcal{O}_K$

original quadruple: [-2, 5*a - 3, a + 3, 6*a - 4]

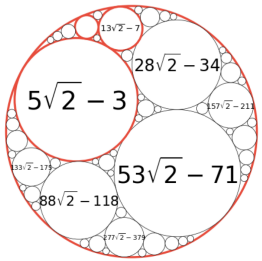
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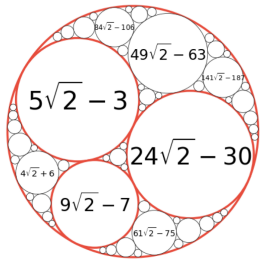
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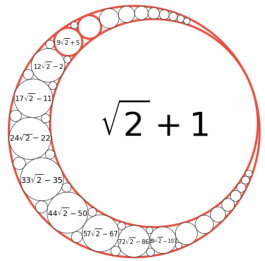
[[-a,1],[-1,0]]



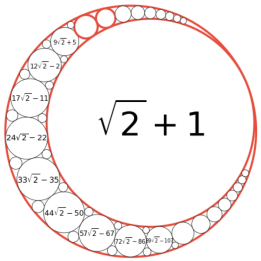
[[-a,-1],[1,0]]



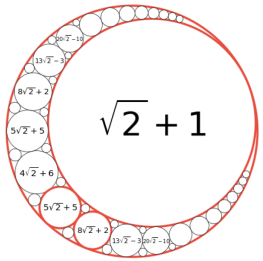
[[-a - 1,0],[-1,-a + 1]]



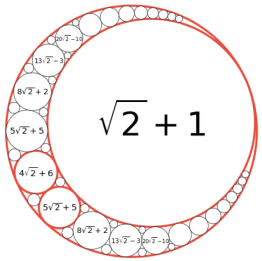
[[-a - 1,0],[-a,-a + 1]]



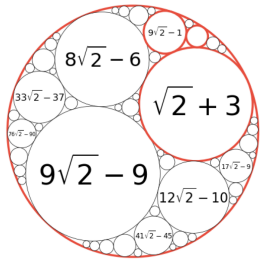
[[-a - 1,0],[a,-a + 1]]



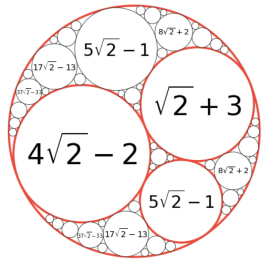
[[-a - 1,0],[1,-a + 1]]



[[-1,0],[-a - 1,-1]]



[[-1,0],[a - 1,-1]]



Thank you!

